DESCRIPTIVE CAPTION GENERATION FOR MANUFACTURING TIME SERIES DATA USING A TIME SERIES LANGUAGE MODEL

A PROJECT REPORT

Submitted by

CHANDIRALEKA G

(2024246032)

A report for the phase-I of the project

submitted to the Faculty of

COMPUTER SCIENCE AND INFORMATION ENGINEERING

in partial fulfillment

for the award of the degree

of

MASTER OF TECHNOLOGY

in

INFORMATION TECHNOLOGY

DEPARTMENT OF INFORMATION SCIENCE AND TECHNOLOGY

COLLEGE OF ENGINEERING, GUINDY

ANNA UNIVERSITY

CHENNAI 600 025

APRIL 2026

ANNA UNIVERSITY

CHENNAI - 600 025

BONAFIDE CERTIFICATE

Certified that this project report titled **DESCRIPTIVE CAPTION GENERATION FOR MANUFACTURING TIME SERIES DATA USING A TIME SERIES LANGUAGE MODEL** is the bonafide work of **CHANDIRALEKA G (2024246032)** who carried out project work under my supervision. Certified further that to the best of my knowledge and belief, the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or an award was conferred on an earlier occasion on this or any other candidate.

PLACE:

DATE:

Dr. P.YOGESH
PROFESSOR
PROJECT GUIDE
DEPARTMENT OF IST, CEG
ANNA UNIVERSITY
CHENNAI 600025

COUNTERSIGNED

Dr. S. SWAMYNATHAN
HEAD OF THE DEPARTMENT
DEPARTMENT OF INFORMATION SCIENCE AND TECHNOLOGY
COLLEGE OF ENGINEERING, GUINDY
ANNA UNIVERSITY
CHENNAI 600025

ABSTRACT

Manufacturing systems generate large volumes of sensor-based time series data, making manual interpretation of equipment behavior and fault patterns both time-consuming and error-prone. Automated time series captioning enhances interpretability by translating complex numerical sensor patterns into natural language descriptions, enabling more effective anomaly detection and condition monitoring.In this work, we adapt the Time Series Language Model (TSLM) for industrial applications. TSLM integrates textual prompts with learned time series representations through a multimodal encoder–decoder architecture, allowing the model to capture subtle temporal dynamics and produce precise, human-readable descriptions of machine behavior.To address the challenge of limited labeled data in industrial environments, the proposed approach employs in-context synthetic data generation followed by cross-modal denoising. This strategy improves the quality and reliability of time series–caption pairs, enabling accurate and robust caption generation. Experimental results demonstrate the effectiveness of the approach for predictive maintenance, supporting early fault detection and improved decision-making in smart manufacturing systems.

CHAPTER 1

INTRODUCTION

Modern manufacturing systems are increasingly instrumented with large numbers of sensors that continuously monitor machine operations, environmental conditions, and production processes. These sensors generate massive volumes of multivariate time series data, capturing valuable information about equipment health, performance trends, and potential faults. However, manually interpreting these complex temporal signals is both time-consuming and challenging, particularly in large-scale industrial environments where rapid decision-making is critical.

Automated analysis of time series data has therefore become a cornerstone of smart manufacturing and Industry 4.0 initiatives. While traditional approaches focus on forecasting and anomaly detection, they typically produce numerical outputs or visual graphs that still require expert interpretation. Transforming raw sensor signals into natural language descriptions offers a more intuitive and human-centered way of understanding system behavior. Automated time series captioning translates salient temporal patterns into textual explanations, improving interpretability, accessibility, and communication of machine conditions across technical and non-technical stakeholders.

Recent advances in Large Language Models (LLMs) have demonstrated remarkable capabilities in understanding and generating natural language across diverse domains. However, their application to industrial time series data remains limited. Most existing LLM-based research in time series has concentrated on forecasting rather than descriptive interpretation.

Furthermore, manufacturing datasets often suffer from data scarcity and limited labeled annotations, making it difficult to directly train language models to produce reliable, domain-specific descriptions of sensor patterns.

To address these challenges, this work adapts the Time Series Language Model (TSLM) for manufacturing time series captioning. TSLM is a multimodal encoder–decoder architecture that integrates textual prompts with learned representations of time series data. By jointly modeling numerical temporal patterns and linguistic context, the model generates precise, human-readable descriptions of equipment behavior. This capability supports predictive maintenance by enabling early identification of abnormal trends and degradation patterns through interpretable textual outputs.

A key challenge in industrial time series captioning is the lack of large-scale annotated datasets. To overcome this limitation, our approach incorporates in-context synthetic data generation using large language models, followed by a cross-modal denoising mechanism that filters noisy or inconsistent time series–caption pairs. This strategy enhances data quality and improves the alignment between sensor patterns and textual descriptions, enabling more robust model training.

Overall, this work bridges the gap between numerical time series analytics and natural language understanding in smart manufacturing. By enabling automated, interpretable descriptions of machine behavior, the proposed approach contributes toward more transparent AI systems, improved fault diagnosis, and more reliable industrial operations

1.1 BACKGROUND

Time series data is fundamental in modern manufacturing systems, where sensors continuously monitor machine operations and production processes. These data streams provide critical insights into equipment health and system performance. However, the increasing volume and complexity of industrial sensor data make manual interpretation difficult and time-consuming.

Traditional time series methods mainly focus on forecasting and anomaly detection, often producing numerical outputs that still require expert analysis. Generating natural language descriptions of time series patterns offers a more intuitive way to interpret machine behavior and communicate insights to engineers and decision-makers.

Although Large Language Models (LLMs) have shown strong capabilities in natural language understanding and generation, they are typically trained on general text and lack knowledge of industrial sensor data. This limits their ability to describe manufacturing time series accurately. Additionally, labeled datasets that pair time series data with textual descriptions are scarce, making supervised training challenging.

To address these issues, this project utilizes a Time Series Language Model (TSLM) that combines learned time series representations with language modeling in a multimodal framework. By incorporating synthetic data generation and cross-modal denoising, the approach improves the alignment between sensor patterns and textual explanations, enabling interpretable monitoring and predictive maintenance in smart manufacturing environments.

1.2 PROBLEM STATEMENT

Manufacturing systems generate large volumes of sensor-based time series data that are difficult to interpret manually due to their complexity and scale.

Existing analysis methods primarily focus on numerical forecasting and anomaly detection, offering limited interpretability and requiring expert knowledge to understand intricate sensor patterns and machine behavior.

Furthermore, labeled datasets that pair manufacturing time series with descriptive captions are scarce, limiting the ability of Large Language Models (LLMs) to learn meaningful mappings between sensor data and natural language explanations.

Therefore, there is a need for an automated approach that can generate accurate and interpretable natural language captions for manufacturing time series, supporting predictive maintenance and improving decision-making in industrial environments.

1.3 OBJECTIVE

The main objective of this project is to develop an automated system capable of generating meaningful natural language descriptions for manufacturing sensor time series data to improve interpretability and support predictive maintenance.. The specific objectives are as follows:

- To design a multimodal Time Series Language Model (TSLM) that integrates numerical time series representations with textual prompts for caption generation.
- To develop a method for synthetic data generation using in-context

prompting to address the scarcity of labeled time series–caption datasets in manufacturing.

- To implement a cross-modal denoising mechanism that improves the quality and alignment of generated time series–caption pairs.
- To generate accurate and interpretable textual descriptions of equipment behavior that assist in early fault detection and condition monitoring.
- To evaluate the effectiveness of the proposed model in supporting predictive maintenance and intelligent decision-making in industrial environments.

1.4 SOLUTION OVERVIEW

The proposes an automated framework for generating interpretable natural language captions from manufacturing sensor time series data. The core of the solution is the Time Series Language Model (TSLM), a multimodal encoder–decoder architecture that integrates numerical time series embeddings with textual prompts to produce descriptive captions.

To overcome the challenge of limited labeled data, the framework employs synthetic data generation through in-context prompting using large language models. The generated captions are then refined using a cross-modal denoising mechanism, which filters out noisy or inconsistent time series–caption pairs by evaluating the alignment between numerical and textual representations.

Once trained, TSLM can process unseen time series data and generate accurate, human-readable captions that describe machine behavior,

detect anomalies, and highlight potential faults. These captions enable engineers and operators to quickly understand system status, facilitating predictive maintenance, anomaly detection, and informed operational decision-making in industrial environments.

The overall solution bridges the gap between raw sensor data and interpretable insights, making time series data more accessible and actionable for manufacturing applications.

CHAPTER 2

LITERATURE SURVEY

This chapter presents a comprehensive review of recent research on Large Language Models (LLMs) for time series analysis, with a particular focus on forecasting, multimodal learning, fault diagnosis, and descriptive interpretation. The goal of this survey is to examine how LLMs are being adapted to numerical temporal data, analyze methodological trends, and identify research gaps that motivate the development of an interpretable time series captioning framework for manufacturing systems.

2.1 LLM-BASED TIME SERIES FORECASTING FRAMEWORKS

Recent research explores adapting LLMs for numerical time series forecasting by transforming temporal signals into language-compatible representations. Unlike traditional statistical models, these approaches leverage pretrained language models to capture temporal dependencies through tokenization, embedding alignment, and prompt-based learning.

Saroj Gopal et al. [1] investigated the use of LLMs to generate executable LSTM forecasting models through prompt-driven synthesis. Their study highlights the feasibility of LLM-assisted model construction but notes sensitivity to prompt design and dataset variability. Chen et al. [2] proposed RTS-LLM, which restores temporal structure by aligning time series signals with language embeddings. While effective for structured datasets, it faces limitations with long-horizon multivariate industrial signals.

Chu et al. [3] introduced Granger-TSLM, integrating Granger

causality into tokenized multivariate forecasting. Although it improves inter-variable dependency modeling, scalability remains challenging for high-dimensional sensor data. Lei et al. [4] presented the CVC framework, which enhances forecasting via cross-view correction and multimodal alignment but increases computational overhead.

Du et al. [5] proposed PMTE-LLM, combining multimodal fusion and reinforcement learning for efficient forecasting. Despite strong predictive performance, the approach depends heavily on domain-specific prompt tuning. Collectively, these studies demonstrate that LLMs can model temporal dynamics but primarily emphasize prediction accuracy over interpretability.

2.2 MULTIMODAL LLMS FOR TIME SERIES UNDERSTANDING

A growing body of research integrates textual knowledge with time series data to enhance contextual reasoning and interpretability. These multimodal systems combine numerical signals with semantic descriptions, domain documentation, or expert annotations.

Wu et al. [6] introduced LGMTime, a zero-shot forecasting model that enriches time series with textual semantic context. The model improves interpretability but depends on high-quality metadata. Zhao et al. [7] proposed a cross-modal knowledge activation framework for zero-shot fault diagnosis, aligning time series signals with textual domain knowledge. However, performance is sensitive to the quality of knowledge alignment.

Zhou et al. [8] developed T2MFD, an LLM-enhanced multimodal fault diagnosis system combining vibration signals and textual diagnostics. While effective, it relies on structured knowledge templates. Moon et al. [9] presented MIFlu, a multimodal influenza forecasting model integrating

contextual text with epidemiological time series, though dependence on external text limits robustness.

These studies show that textual augmentation enhances reasoning but often requires structured domain knowledge not always available in industrial settings.

2.3 LLMS FOR INDUSTRIAL FAULT DIAGNOSIS

Several works apply LLMs to industrial monitoring and fault detection tasks, where interpretability and reliability are critical. These approaches attempt to bridge numerical sensor analysis with human-understandable explanations.

Zhang et al. [10] proposed LLM-TSFD, a human-in-the-loop fault diagnosis system that supports interpretable decision-making. While effective, scalability is limited due to expert involvement. Lin et al. [11] explored symbolic representation learning for time series classification using BERT-based encoders, showing that discretized signals can be processed by language models, though performance depends on symbolization strategies.

TempoGPT by Zhang et al. [12] quantizes time series into tokens for reasoning in a shared embedding space. Despite strong reasoning ability, it depends on predefined codebooks and structured assumptions. These works demonstrate LLM potential in industrial diagnostics but focus more on classification and reasoning than descriptive explanation.

2.4 DOMAIN-SPECIFIC LLM APPLICATIONS IN TIME SERIES

LLMs have been adapted for specialized temporal domains where contextual knowledge plays a key role. Chaffard et al. [13] enhanced LLMs for Bitcoin time series forecasting using financial domain knowledge. Although accurate in finance, transferability to other industries is limited.

Xu et al. [14] proposed a dual-LLM framework for maritime greenhouse gas emission forecasting using multi-source temporal data. While effective, the model incurs high computational cost. Wen et al. [15] applied LLM-based temporal description alignment for remote sensing crop classification, but performance declines under noisy or irregular signals.

These domain-focused systems highlight adaptability but reveal limited cross-domain generalization.

2.5 TIME SERIES LANGUAGE MODELING AND CAPTIONING

Among emerging approaches, Trabelsi et al. [16] introduced the Time Series Language Model (TSLM), designed to generate natural language descriptions from time series inputs. Their multimodal encoder-decoder integrates textual prompts with numerical embeddings and addresses data scarcity using synthetic data and cross-modal denoising. While promising for interpretability, evaluation was limited to smaller datasets, leaving open challenges in large-scale industrial applications.

2.6 RESEARCH GAPS

From the reviewed literature, several research gaps emerge:

- Most LLM-based time series studies focus on forecasting and

classification rather than natural language captioning.

- Many approaches rely on structured metadata, handcrafted prompts, or expert feedback, reducing scalability in real-world manufacturing environments.
- Multimodal methods often struggle with noisy, high-dimensional sensor streams and limited labeled data.
- Interpretability remains secondary to predictive accuracy in most industrial LLM systems.

2.7 SUMMARY

This chapter reviewed recent advancements in LLM-based time series forecasting, multimodal learning, industrial diagnostics, and domain-specific applications. While LLMs show strong potential for temporal reasoning, current research largely emphasizes prediction and classification rather than interpretable natural language description. Challenges related to scalability, data scarcity, and domain adaptation persist. These limitations motivate the development of an automated and interpretable time series captioning framework for manufacturing environments, which forms the focus of this work.

CHAPTER 3

SYSTEM DESIGN AND IMPLEMENTATION

3.1 System Architecture

The proposed system is an automated **Time Series Language Model (TSLM)** framework designed for large-scale time series captioning in manufacturing and industrial monitoring applications. As shown in the system architecture diagram, the framework integrates synthetic data generation, time series representation learning, cross-modal denoising, and transformer-based caption generation into a unified end-to-end pipeline.

The system analyzes raw time series sensor data and generates descriptive, human-understandable natural language captions that summarize system behavior, operational conditions, and potential anomalies. The architecture is modular, scalable, and designed to operate efficiently even under limited labeled data availability.

The architecture consists of multiple interconnected modules, starting from ground-truth dataset preparation and synthetic data generation, followed by time series encoding, denoising via cross-modal retrieval, and final caption generation using TSLM.

The major modules implemented in this architecture are listed below:

- Ground-Truth TS–Caption Dataset Module
- Synthetic Data Generation Module

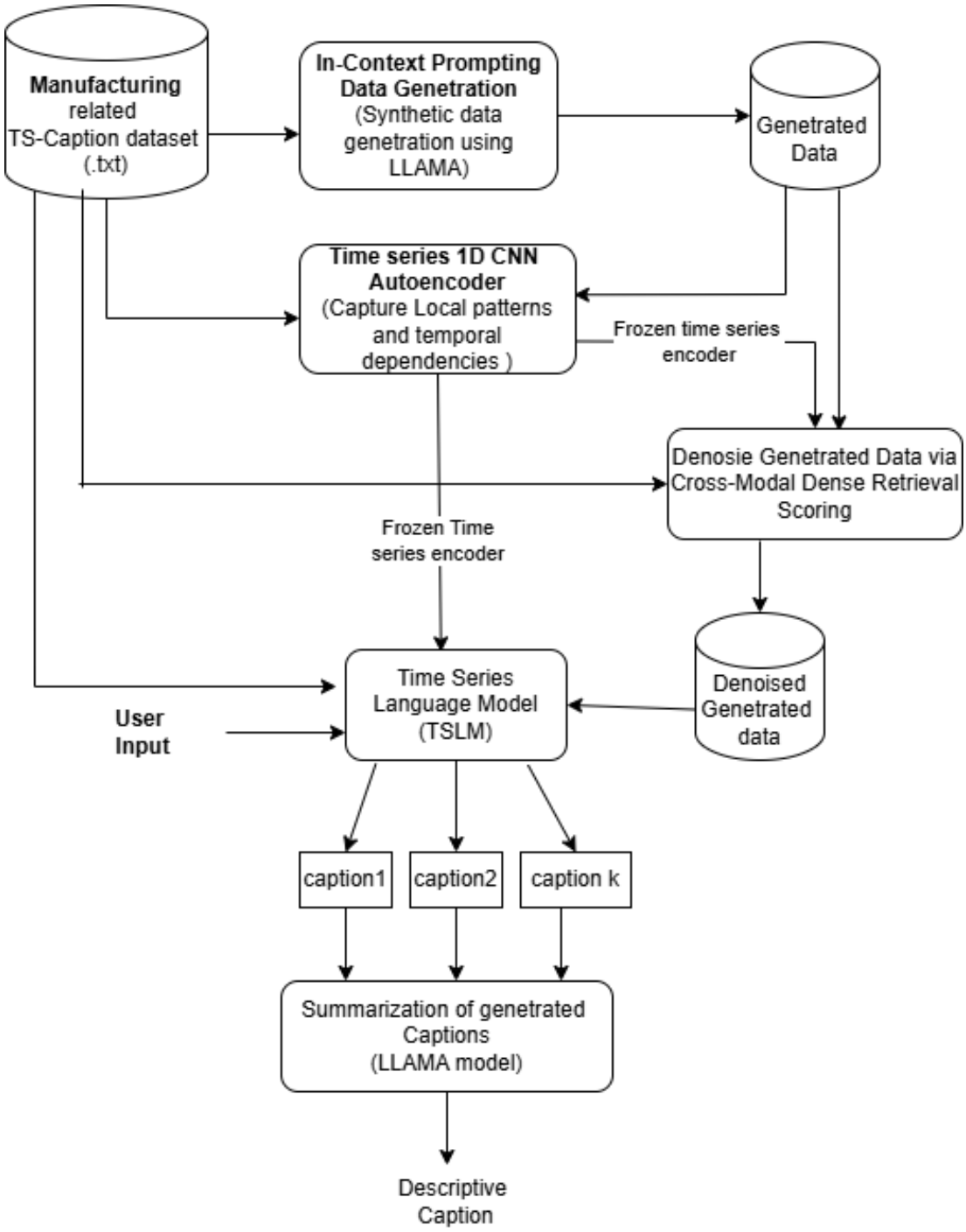

Figure 3.1: Time Series Language Model (TSLM) Architecture

- Time Series Representation Learning Module
- Cross-Modal Denoising Module
- Time Series Language Model (TSLM) Module

- Caption Summarization Module

3.2 Ground-Truth TS–Caption Dataset Module

The ground-truth dataset module forms the foundation of the proposed system. This dataset consists of paired time series sensor readings and corresponding human-written captions describing system conditions. The time series data are stored in numerical format, while the captions are stored as textual descriptions.

The dataset represents various operational states of manufacturing and hydraulic systems, including normal operation, degradation, and fault conditions. This ground-truth dataset is used for few-shot prompting, representation learning, and training the denoising and captioning models.

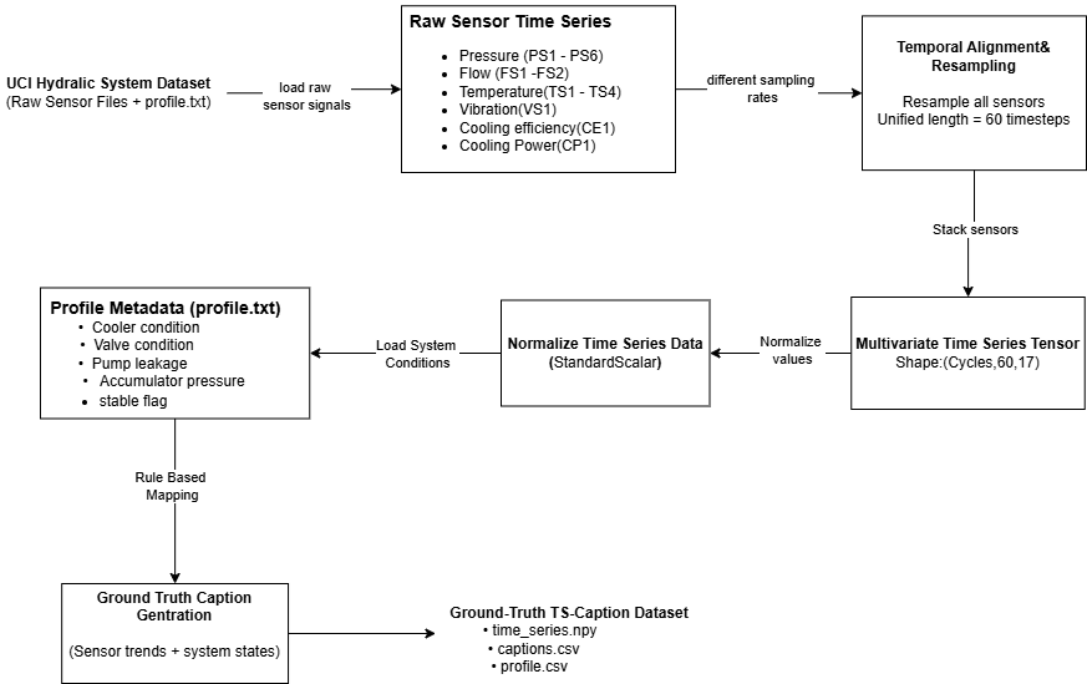

Figure 3.2: Module 0: Ground-Truth TS–Caption Dataset

3.3 InContext-Prompting Data Generation Module

The synthetic data generation module, shown in Fig. 3.3, addresses the problem of labeled data scarcity in time series captioning. This module leverages an open-source Large Language Model (LLM) such as LLaMA or Mistral to generate large-scale synthetic time series–caption pairs.

Few-shot demonstrations are selected from the ground-truth dataset and converted into tagged textual representations using temporal segments such as <BEGIN>, <MIDDLE>, and <END>. These demonstrations are used as in-context examples to guide the LLM in generating realistic synthetic data. The generated samples are stored incrementally in JSONL format.

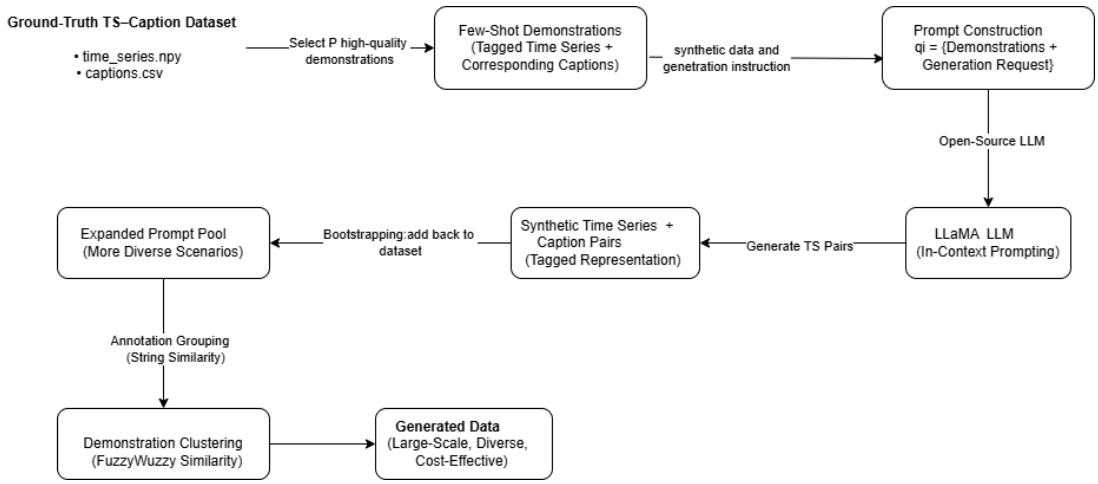

Figure 3.3: Module 1: Synthetic Data Generation using In-Context Prompting

3.4 Time Series Representation Learning Module

The time series representation learning module, shown in Fig. 3.4, employs a 1D Convolutional Neural Network (CNN) based autoencoder to learn latent representations of time series data.

The encoder captures local temporal patterns and dependencies using stacked 1D convolution layers with down-sampling. A linear projection layer maps the extracted features into a fixed-dimensional embedding space aligned

with text embeddings. The decoder reconstructs the input time series during training using up-sampling and deconvolution layers.

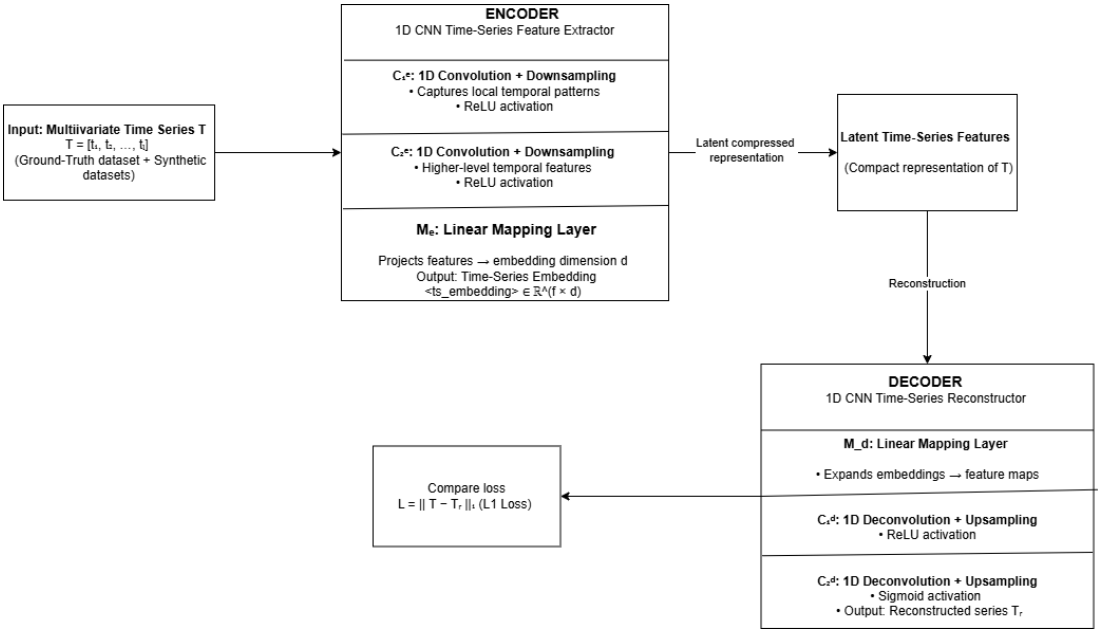

Figure 3.4: Module 2: Time Series 1D CNN Autoencoder

After training, only the encoder is retained and frozen. These frozen embeddings are used in subsequent modules to ensure stable and efficient representation learning.

3.5 Cross-Modal Denoising Module

The cross-modal denoising module, illustrated in Fig. 3.5, filters noisy and low-quality synthetic data using dense retrieval scoring. A multi-modal encoder projects both time series embeddings and caption embeddings into a shared embedding space.

The model is trained using contrastive learning with in-batch negatives, ensuring that semantically aligned time series–caption pairs achieve higher similarity scores. After training, similarity scores are computed for all

generated samples, and pairs with similarity below a predefined threshold are removed.

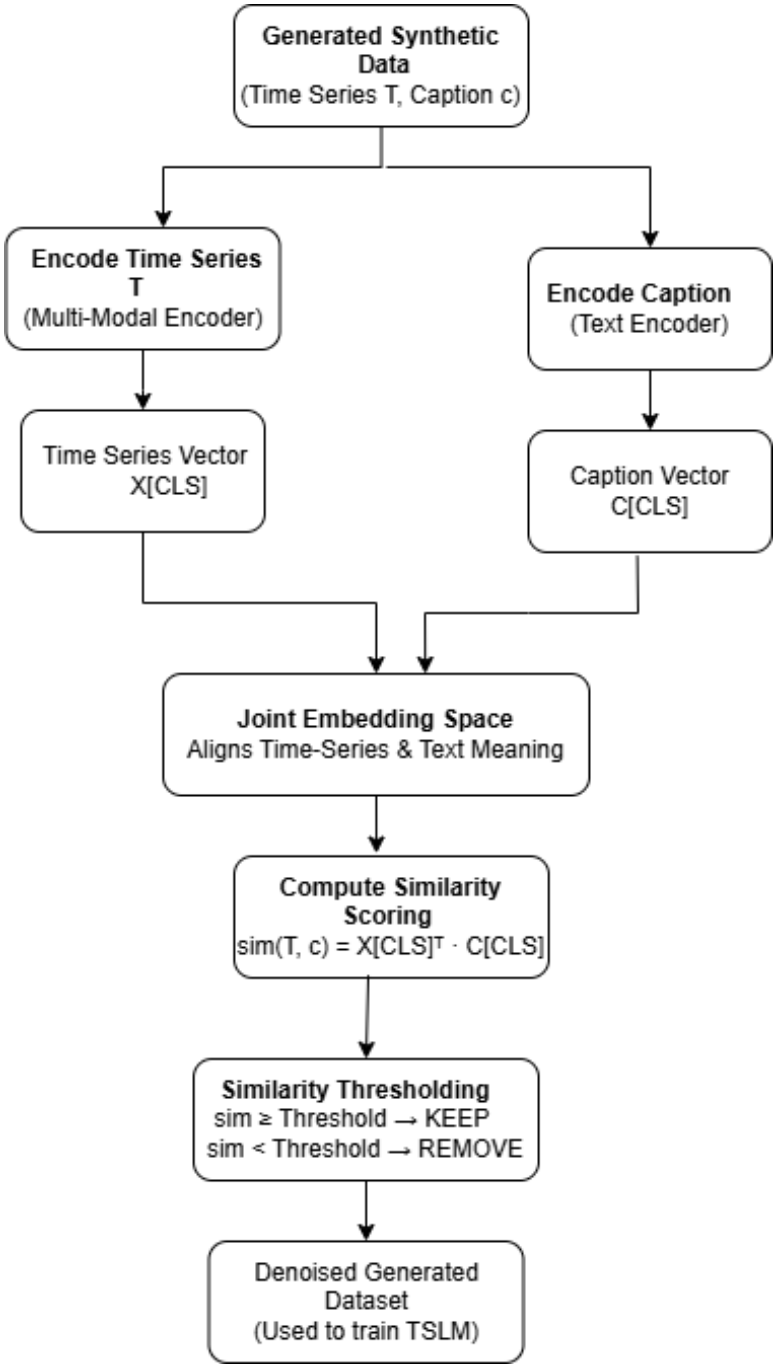

Figure 3.5: Module 3: Cross-Modal Dense Retrieval-Based Denoising

The output of this module is a large-scale, high-quality denoised synthetic dataset used for training the TSLM.

3.6 Time Series Language Model (TSLM) Module

The TSLM module, shown in Fig. 3.6, is responsible for generating natural language captions from time series data. The model employs a multi-modal encoder to generate joint representations of time series embeddings and textual tokens.

A transformer-based text decoder with self-attention and cross-attention layers generates captions in an autoregressive manner. The model is trained using both the ground-truth dataset and the denoised synthetic dataset through next-token prediction with teacher forcing.

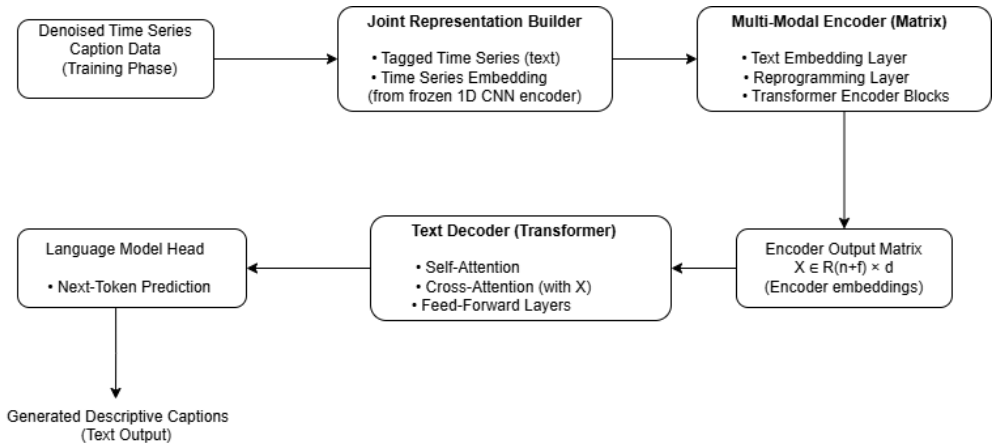

Figure 3.6: Module 4: Time Series Language Model (TSLM)

3.7 Caption Summarization Module

The caption summarization module aggregates multiple captions generated by TSLM into a single concise and descriptive summary. An LLM-based summarization model is used to combine diverse captions into a coherent final description.

This module improves readability and ensures that the final output is easily interpretable by human users, particularly for industrial monitoring and

decision-support applications.

3.8 Summary

This chapter presented the system design and implementation of the proposed Time Series Language Model framework. The architecture integrates synthetic data generation, time series representation learning, cross-modal denoising, and transformer-based caption generation into a unified pipeline. By combining deep learning with large-scale synthetic data and automatic denoising, the proposed system enables scalable, accurate, and interpretable time series captioning for industrial applications.

REFERENCES

10

[1] Saroj Gopal et al. Prompt-driven synthesis of lstm forecasting models using large language models. *Expert Systems with Applications*, 244:122978, 2024.

[2] X. Chen et al. RTS-LLM: Restoring temporal structure for time series forecasting with large language models. *Knowledge-Based Systems*, 290:111456, 2024.

[3] Y. Chu et al. Granger-tslm: Integrating granger causality into tokenized large language models for multivariate time series forecasting. *Information Fusion*, 104:102210, 2024.

[4] Z. Lei et al. CVC: Cross-view correction framework for multimodal time series forecasting using large language models. *Pattern Recognition*, 155:110789, 2024.

[5] H. Du et al. PMTE-LLM: Reinforcement learning-enhanced multimodal large language model for time series forecasting. *Engineering Applications of Artificial Intelligence*, 134:108621, 2024.

[6] T. Wu et al. LGMTime: Large generative models for zero-shot time series forecasting with textual context. *Proceedings of the AAAI Conference on Artificial Intelligence*, 38(16):17845–17853, 2024.

[7] L. Zhao et al. Cross-modal knowledge activation for zero-shot industrial fault diagnosis using large language models. *Mechanical Systems and Signal Processing*, 205:110987, 2024.

[8] J. Zhou et al. T2MFD: Text-to-multimodal fusion diagnosis framework powered by large language models. *Reliability Engineering and System Safety*, 247:110254, 2024.

[9] S. Moon et al. MIFlu: Multimodal influenza forecasting with large language models and temporal signals. *IEEE Journal of Biomedical and Health Informatics*, 28(6):3452–3463, 2024.

[10] K. Zhang et al. LLM-TSFD: Human-in-the-loop large language model framework for interpretable time series fault diagnosis. *IEEE Transactions on Industrial Informatics*, 20(5):4123–4134, 2024.

[11] D. Lin et al. Symbolic representation learning for time series classification using transformer-based language models. *Pattern Recognition Letters*, 182:89–97, 2024.

[12] Y. Zhang et al. TempoGPT: Tokenized time series reasoning with large language models. *Neurocomputing*, 574:127183, 2024.

[13] Q. Chaffard et al. Enhancing large language models for cryptocurrency time series forecasting using financial domain knowledge. *Expert Systems with Applications*, 246:123145, 2024.

[14] M. Xu et al. Dual large language model framework for maritime greenhouse gas emission forecasting. *Applied Energy*, 362:122995, 2024.

[15] C. Wen et al. Large language model assisted temporal description alignment for remote sensing crop classification. *ISPRS Journal of Photogrammetry and Remote Sensing*, 210:145–158, 2024.

[16] A. Trabelsi et al. TSLM: Time series language model for automatic caption generation of temporal data. *Information Fusion*, 106:102318, 2024.